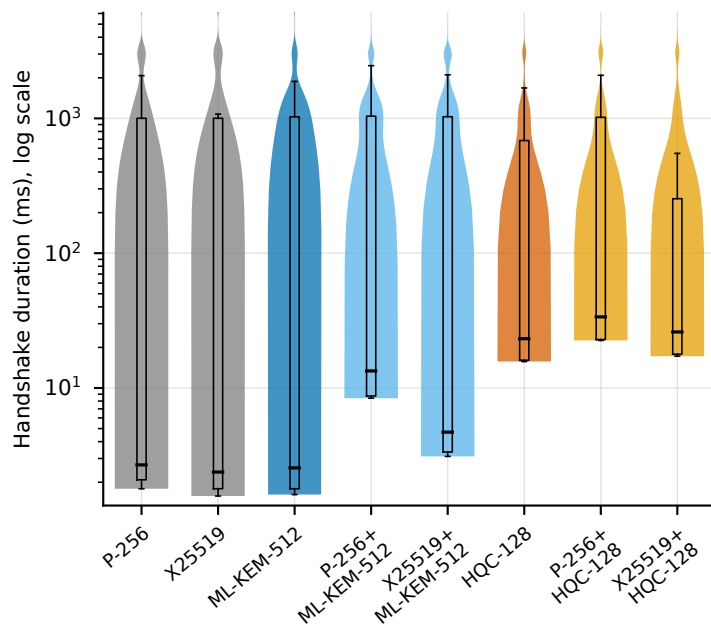


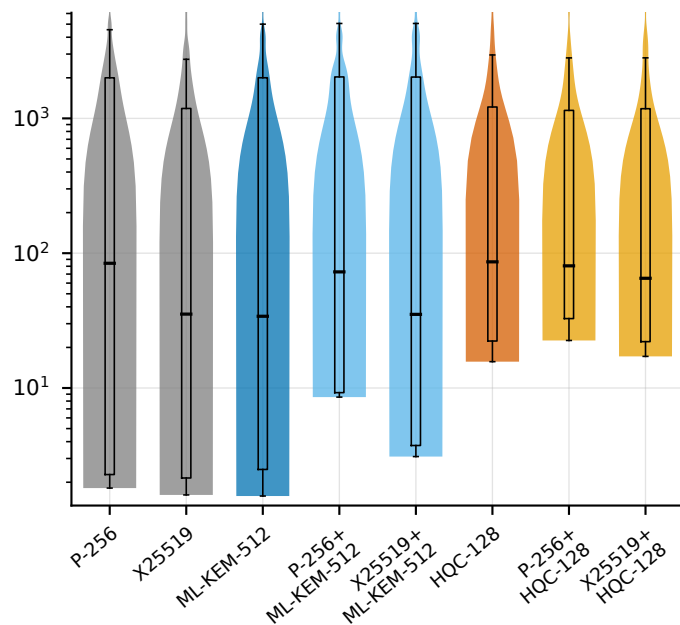
(a)

ed25519 (classical)



(b)

mldsa44 (post-quantum)



(c)

Classical KEX ML-KEM (pure) ML-KEM (hybrid) HQC (pure) HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	ed25519 Fail	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)	mldsa44 Fail
P-256	917.4	194.6	10.00k	25/500	1.28k	171.0	15.50k	8/500
X25519	1.06k	189.4	10.59k	21/500	1.13k	175.2	11.74k	19/500
ML-KEM-512	901.7	198.6	14.45k	14/500	1.23k	171.1	12.94k	16/500
P-256+ML-KEM-512	732.3	176.1	7.56k	13/500	1.29k	162.1	12.88k	8/500
X25519+ML-KEM-512	749.4	184.2	10.54k	10/500	1.41k	165.2	14.19k	13/500
HQC-128	523.1	221.6	9.31k	3/500	1.24k	195.1	23.21k	8/500
P-256+HQC-128	585.0	213.8	9.10k	4/500	1.11k	189.0	15.08k	6/500
X25519+HQC-128	575.0	230.6	8.75k	5/500	1.05k	185.9	11.66k	7/500